

ECSE-4040

Digital Electronics

Lecture 24

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Lecture Outline

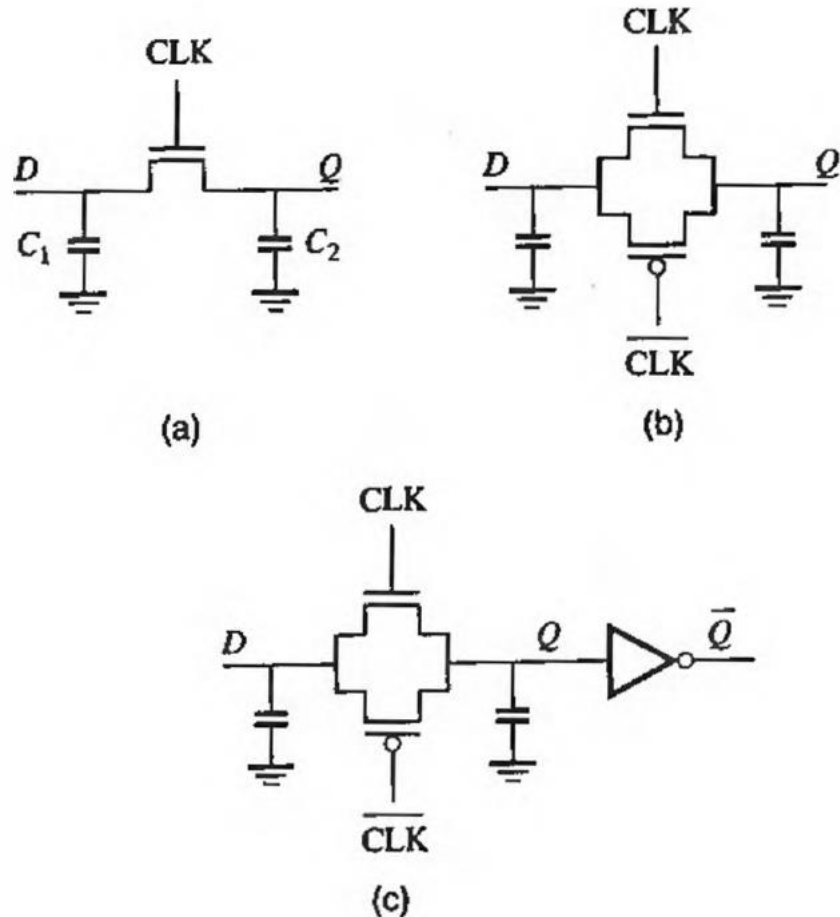
- Dynamic D Latches and D FFs
- Dynamic Logic
- Domino Logic

Progression of Simple D Latches

Problems:

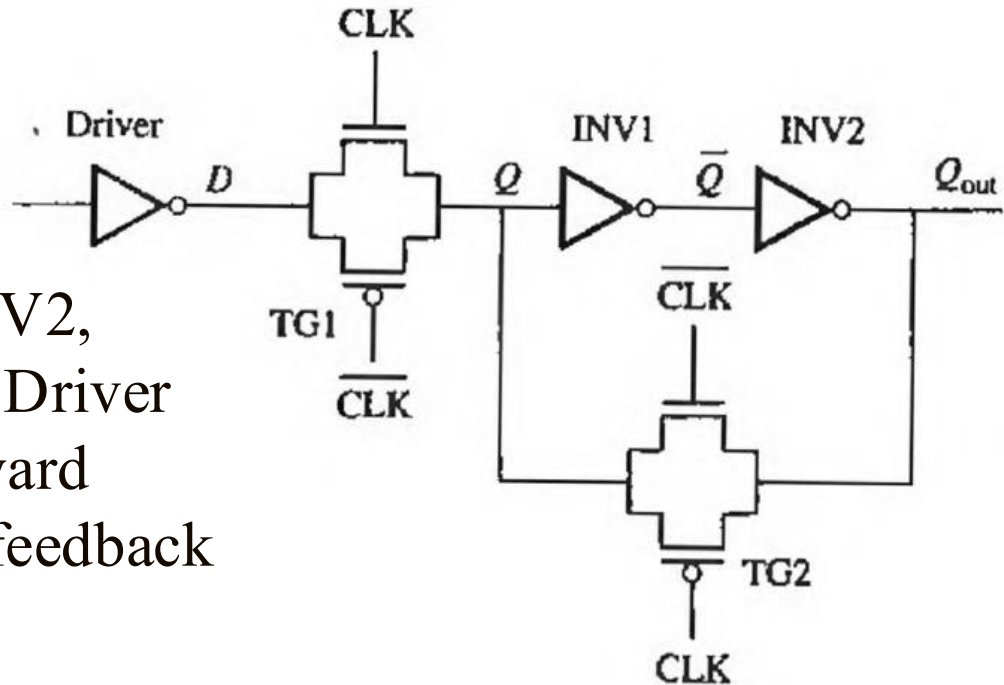
- V_T cutoff
- Clock feedthrough
- No $\sim Q$ output
- Output at high Z when clock goes low

Dynamic storage at Q is a problem!

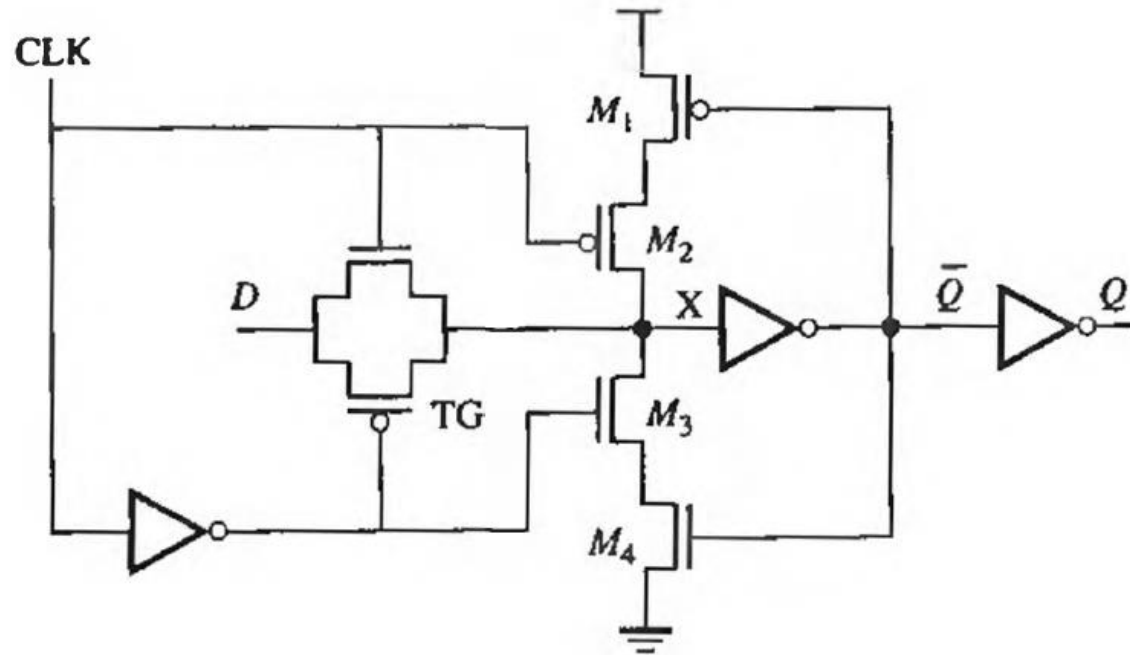


CMOS D Latches using TGs

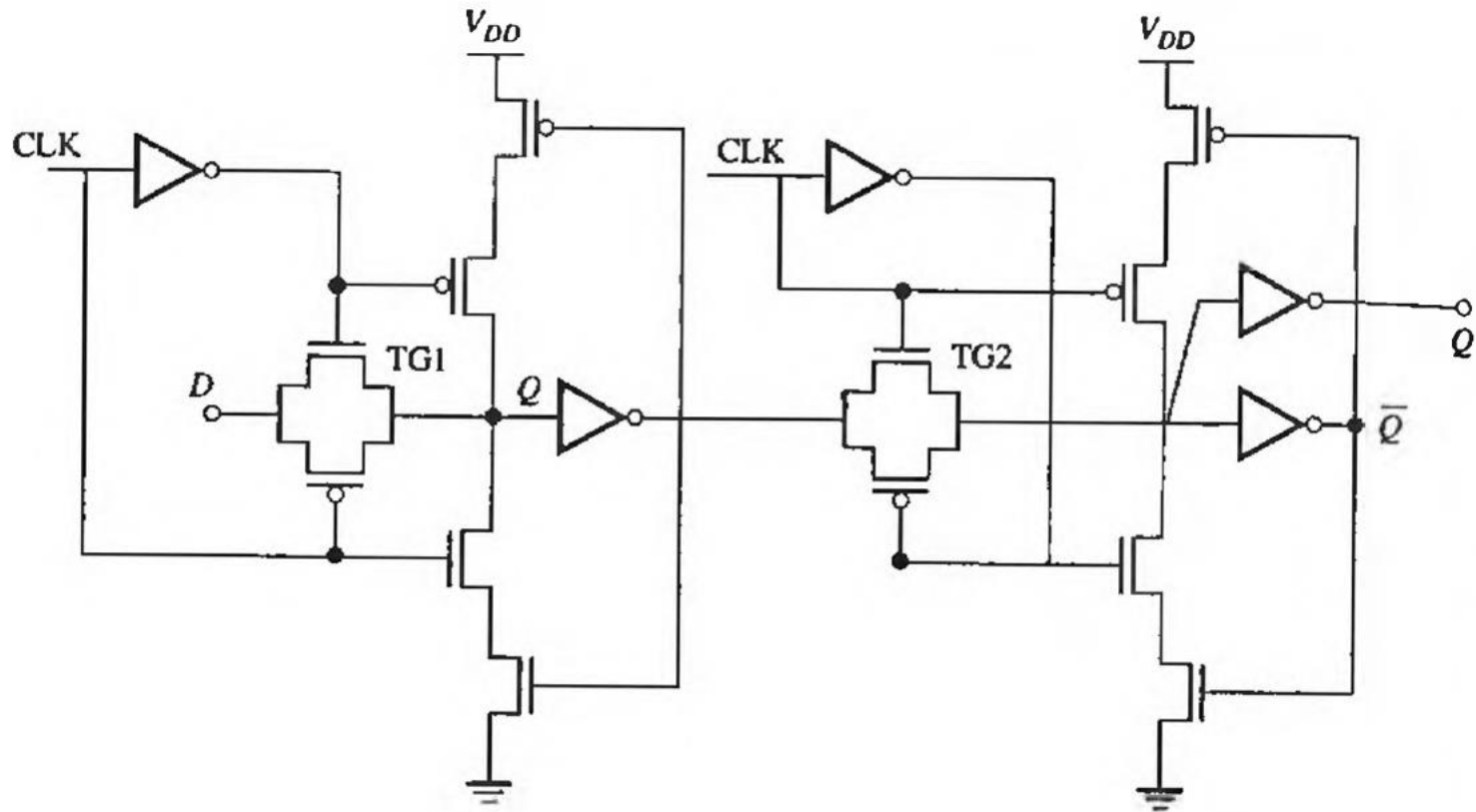
Proper sizing of INV2, TG2 and TG1, and Driver to D, ensuring forward path stronger than feedback path.



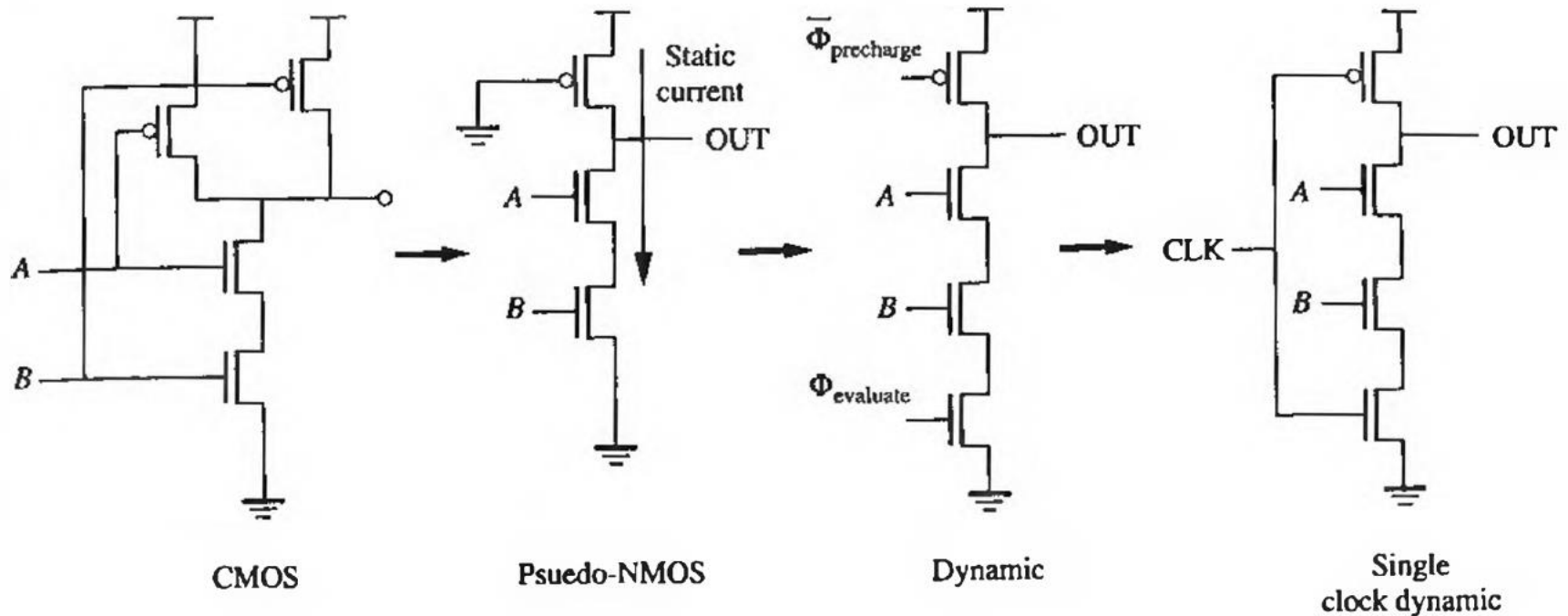
Typical D Latch in CMOS



Positive Edge-Triggered D FF

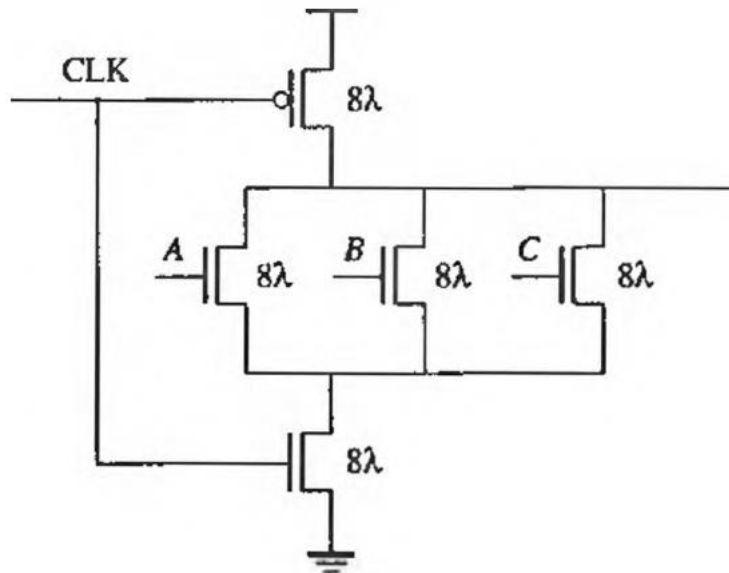


Static to Dynamic Gates

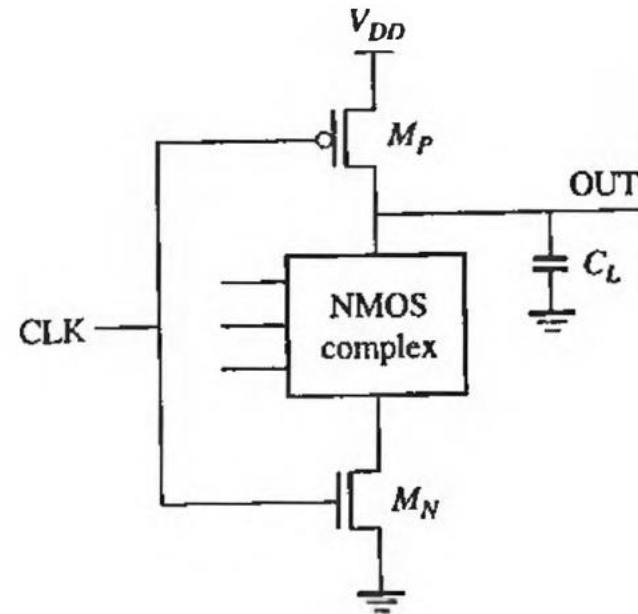


Dynamic Gates

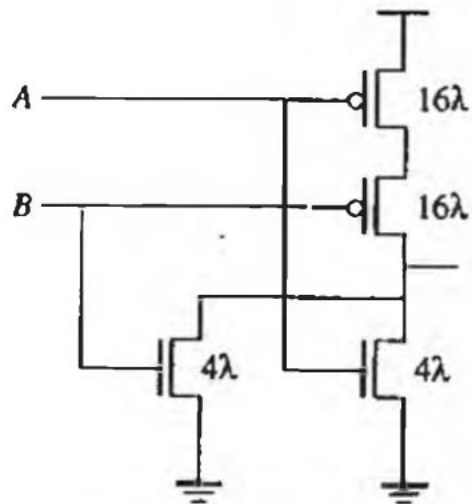
3-input NOR Gate



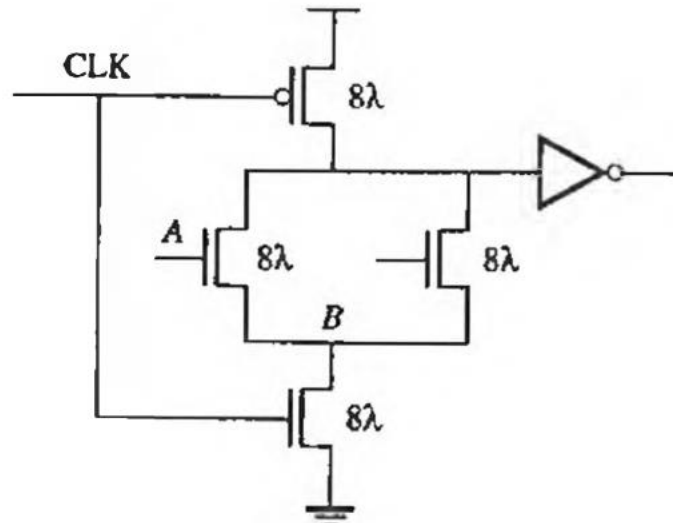
General Structure



Static vs. Dynamic Logic

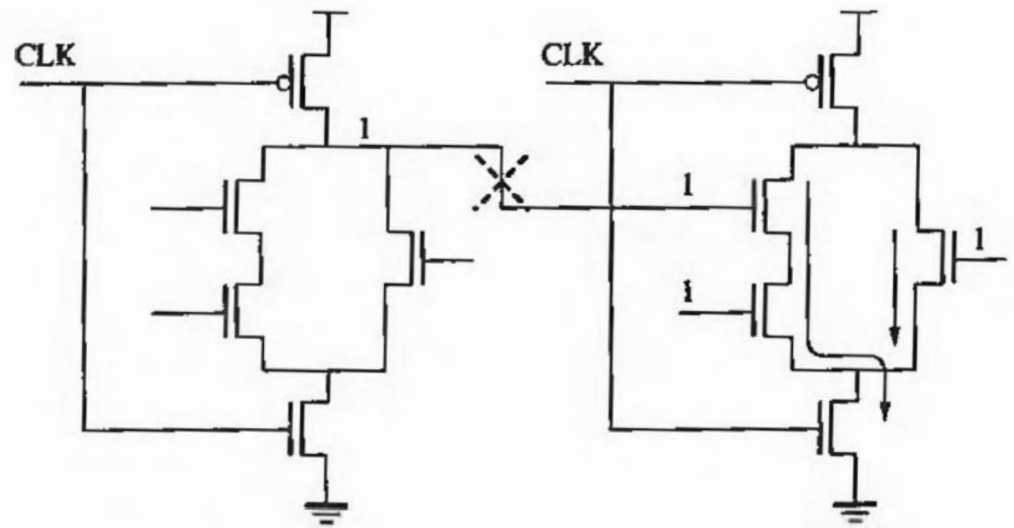
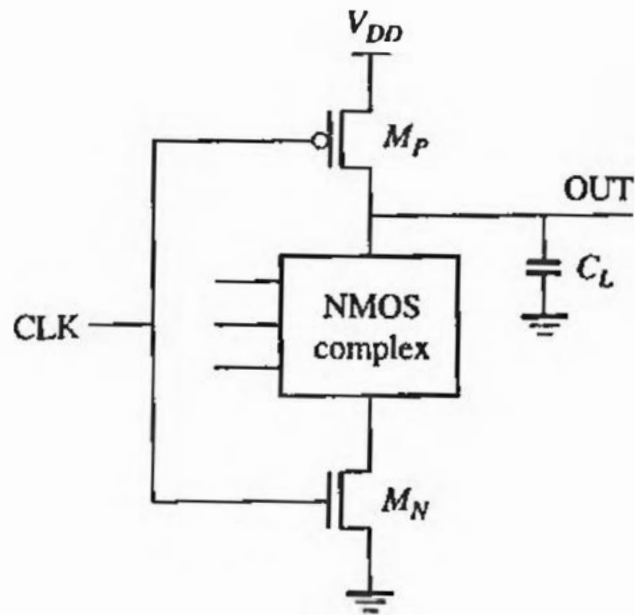


(a) Static NOR gate



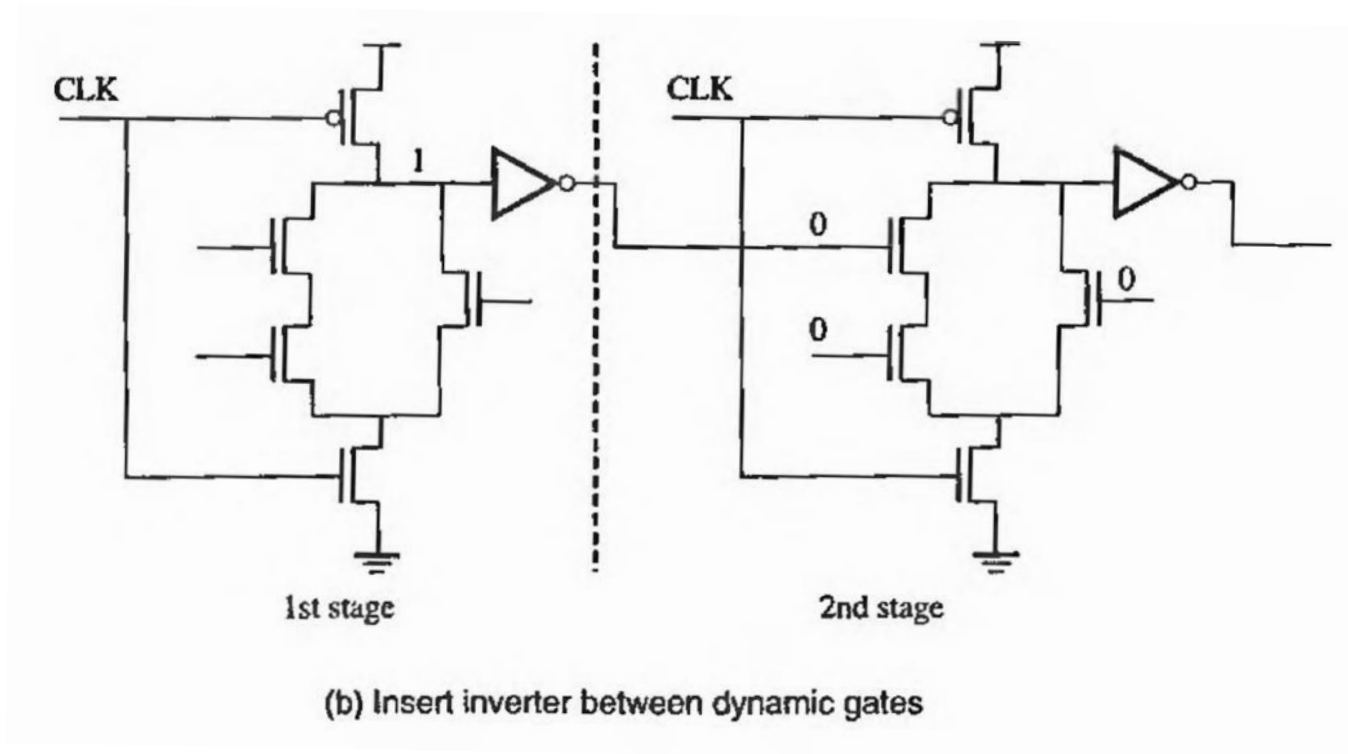
(b) Dynamic NOR gate

Dynamic Gates

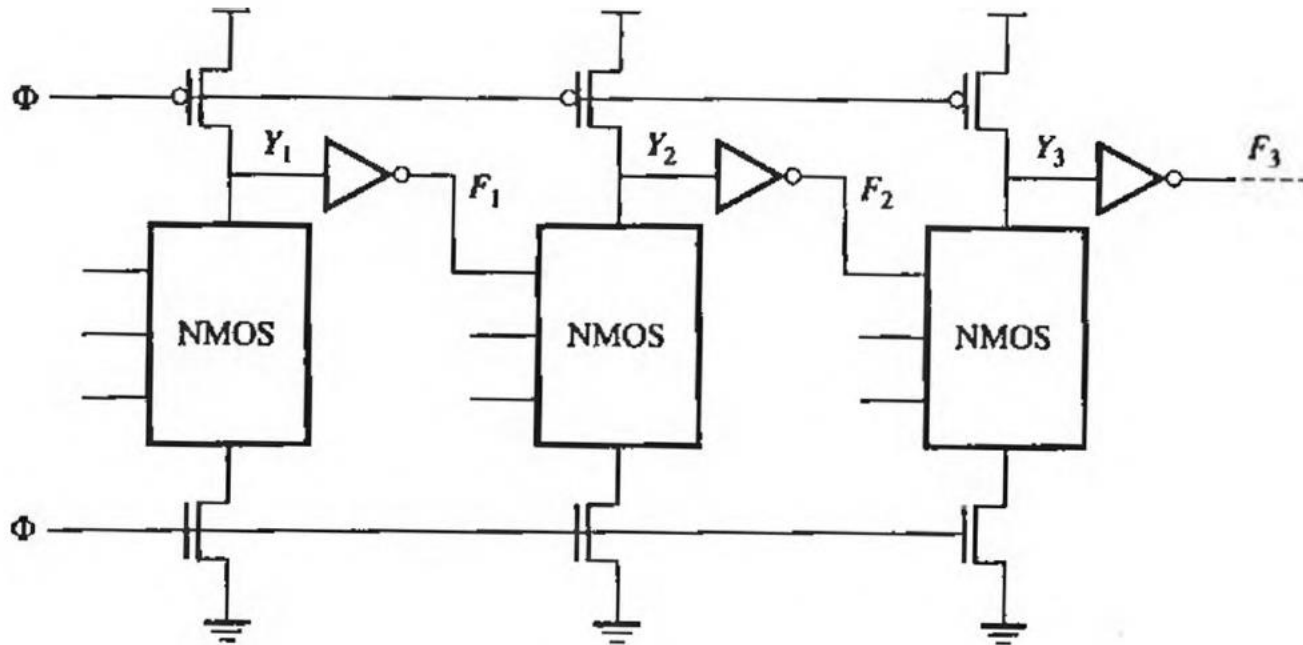


(a) Direct connection not possible

Dynamic Gates



Domino Logic Stages



Domino Logic

- In evaluation phase, Φ goes high and internal nodes (Y_1 , Y_2 , Y_3) will fall like dominos, if path to gnd exists.
- True outputs F_1 , F_2 , F_3 would rise to V_{DD} , dependent on n-complexes. Clock signal needs to be long enough for logic to propagate entire chain-asymmetric clock with high duty cycle.
- Stronger pull-down in dynamic gate and strong pull-up in static inverter, causing a *skewed* inverter and lower switching voltage, V_S .

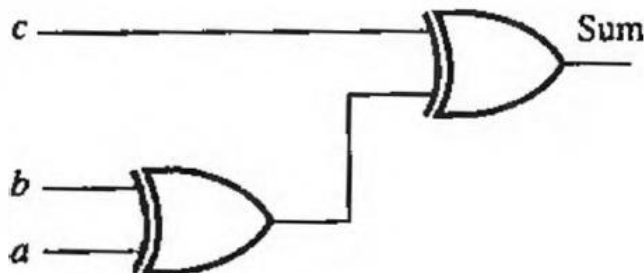
Adder (HJS Ex. 7.7)

Problem:

Implement the function $sum = a \oplus b \oplus c$ in domino logic. Assume that the literals $a, \bar{a}, b, \bar{b}$ are available as stable inputs to the gates.

Solution:

We need to create the XOR and XNOR of a and b using domino gates and then apply them to a third XOR gate. Note that we need the true and complement of X to implement the function.

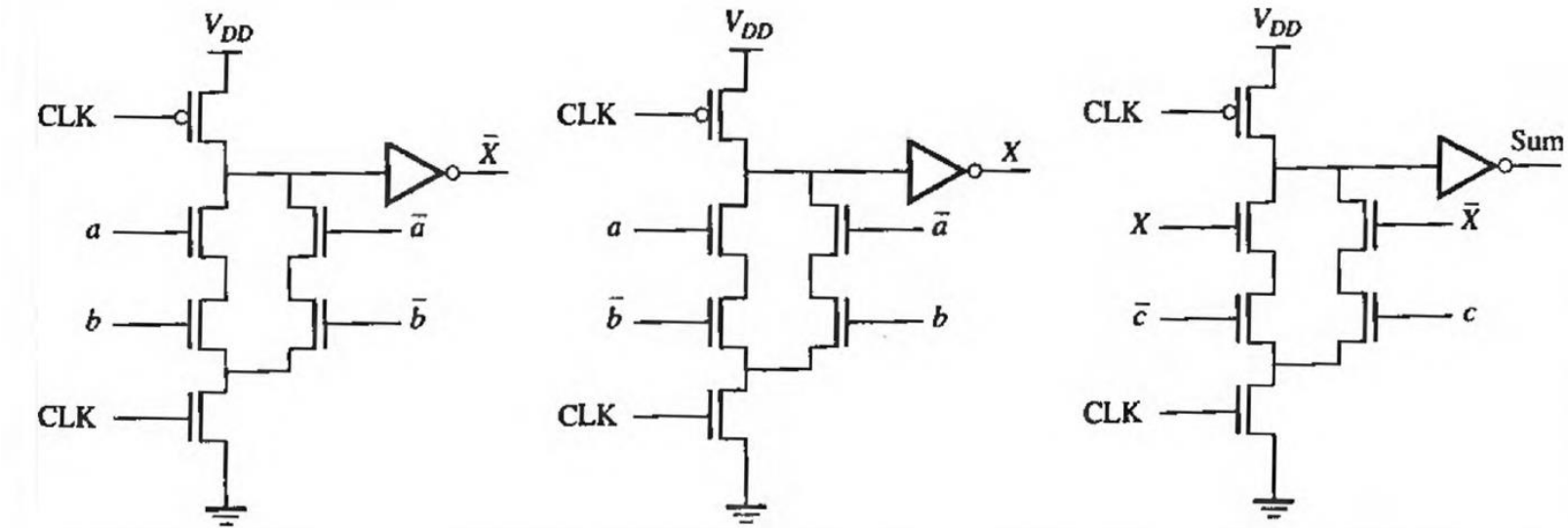


$$X = a \oplus b = a\bar{b} + \bar{a}b$$

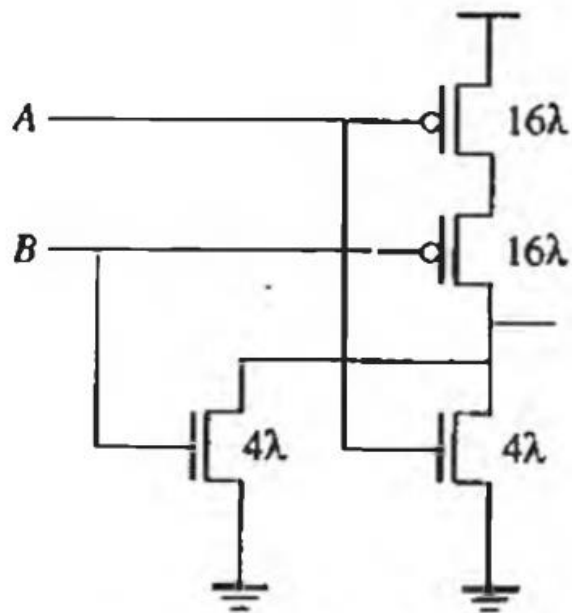
$$\bar{X} = \overline{a \oplus b} = ab + \bar{a}\bar{b}$$

$$Sum = c \oplus X = c\bar{X} + \bar{c}X$$

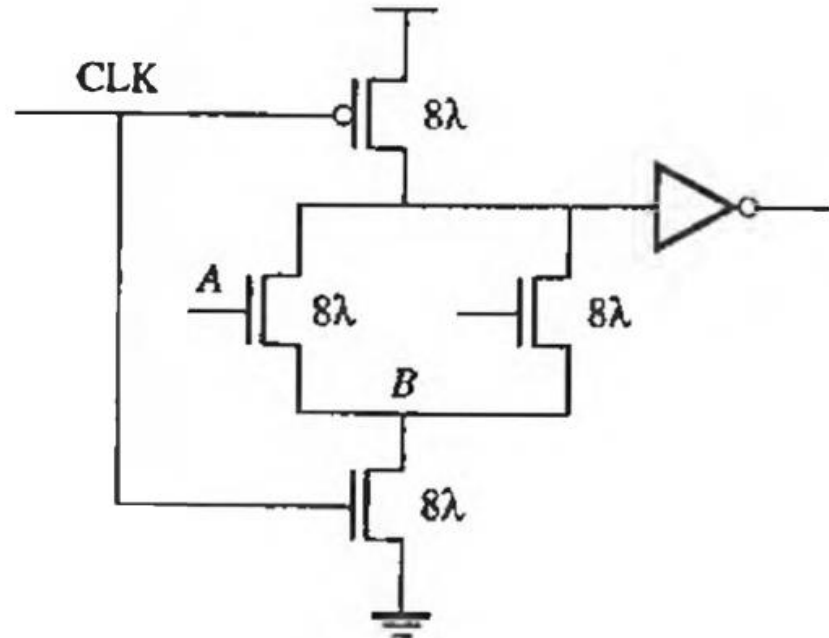
Adder (HJS Ex. 7.7)



Static vs. Dynamic Logical Efforts



(a) Static NOR gate



(b) Dynamic NOR gate

Static vs. Dynamic Logical Efforts

inverter. The inverter size of 8λ for the pull-up and 4λ for the pull-down produces a ratio of

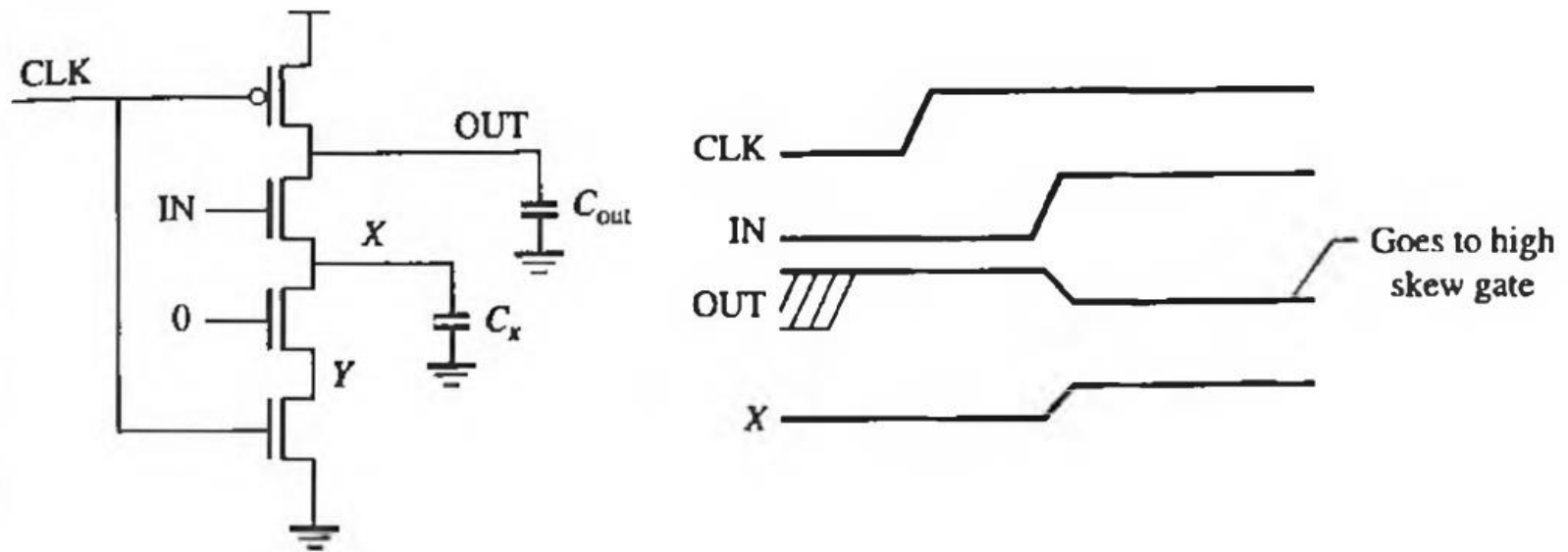
$$LE_{\text{NOR}} = \frac{(\text{NOR input cap.})}{(\text{Inverter input cap.})} = \frac{16\lambda + 4\lambda}{8\lambda + 4\lambda} = 5/3$$

For the domino circuit, the input capacitance is simply 8λ since it is only connected to NMOS devices. Then,

$$LE_{\text{dyn_NOR}} = \frac{(\text{Dynamic input cap.})}{(\text{Inverter input cap.})} = \frac{8\lambda}{12\lambda} = 2/3$$

Since the extra inverter in domino logic has an $LE = 1$, the average LE for each of the two gates is the geometric mean of the product of their logical efforts (i.e., $LE_{\text{avg}} = \sqrt{(2/3)(1)} \approx 0.8$). Therefore, the domino stage is better in terms of its overall drive capability and input capacitive loading.

Limitations of Domino Logic



Charge sharing between OUT and X nodes

$$V^* = \frac{C_{out}}{C_x + C_{out}} V_{DD} \sim V_{DD}/2$$

Minimize Charge Sharing

- Increase output capacitance
- Precharge internal node X to V_{DD} during precharge phase with additional transistor
- Introduce “keepers” or “baby-sitters” to hold output value high in presence of charge-sharing

Limitations of Domino Logic

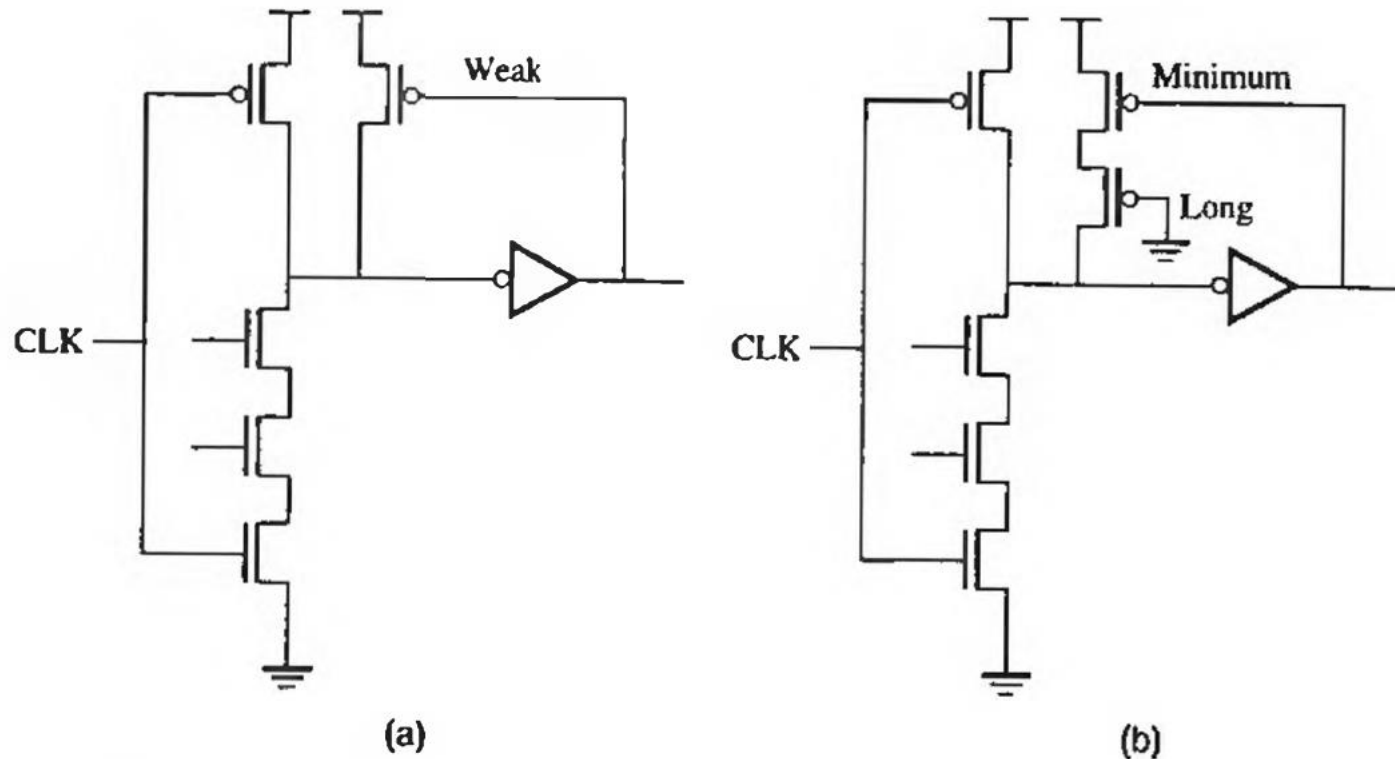


Figure 7.31

Minimizing the effects of charge sharing using keepers.

Other Issues with Domino Logic

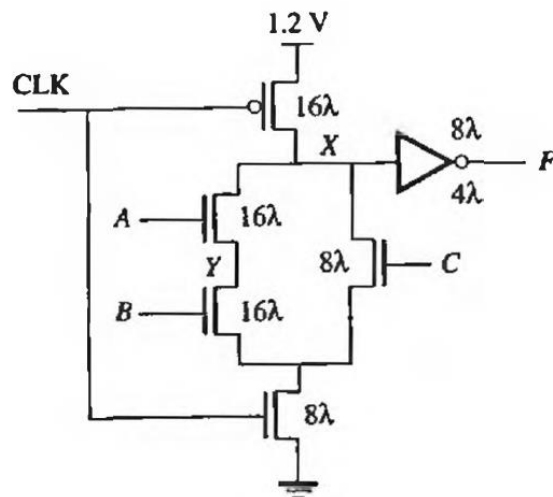
- Leakage
 - Junction and subthreshold leakage
 - Alpha particles
- Noise injection due to capacitive coupling
 - Clock feedthrough through C_{GD} capacitance of PMOS device

Charge Sharing and Feedthrough (HJS Ex. 7.8)

Problem:

For the domino function below, assume $0.13\ \mu\text{m}$ technology parameters and answer the following questions:

- (a) What function does the gate perform at the output F ?
- (b) How much clock feedthrough do we observe at the internal node X ? Is this a potential problem?
- (c) What is the worst-case charge sharing voltage that we observe at node X ?



Charge Sharing and Feedthrough (HJS Ex. 7.8)

Solution:

- (a) The function at the output is $F = AB + C$ (a noninverting function).
- (b) Clock feedthrough is due to the clock going from low to high. The PMOS device is in the linear region after precharge. The feedthrough capacitance is

$$C_f = C_{GD} = \frac{C_g W_p}{2} = \frac{(2 \text{ fF}/\mu\text{m})(16)(0.05 \mu\text{m})}{2} = 0.8 \text{ fF}$$

The grounded capacitance at node X, assuming no source/drain sharing, is

$$\begin{aligned} C_{\text{gnd}} &= C_{\text{eff}}(16\lambda + 16\lambda + 8\lambda) + C_g(8\lambda + 4\lambda) \\ &= (1 \text{ fF}/\mu\text{m})(40)(0.05 \mu\text{m}) + (2 \text{ fF}/\mu\text{m})(12)(0.05 \mu\text{m}) \\ &= 5 \text{ fF} \end{aligned}$$

The feedthrough voltage is

$$\Delta V_x = \frac{C_f \Delta V_{\text{clk}}}{C_f + C_{\text{gnd}}} = \frac{0.8 \text{ fF}(1.2 \text{ V})}{0.8 \text{ fF} + 5 \text{ fF}} = 0.17 \text{ V}$$

Therefore the output voltage due to feedthrough is $1.2 + 0.17 \text{ V} = 1.37 \text{ V}$.
Since the value is above V_{DD} , it is not a problem.

Charge Sharing and Feedthrough (HJS Ex. 7.8)

- (c) The worst-case charge sharing assumes that the output is sitting at 1.2 V, $A = 1, B = C = 0$. We have already determined that $C_X = 5$ fF. The capacitance at the intermediate node between transistors A and B is

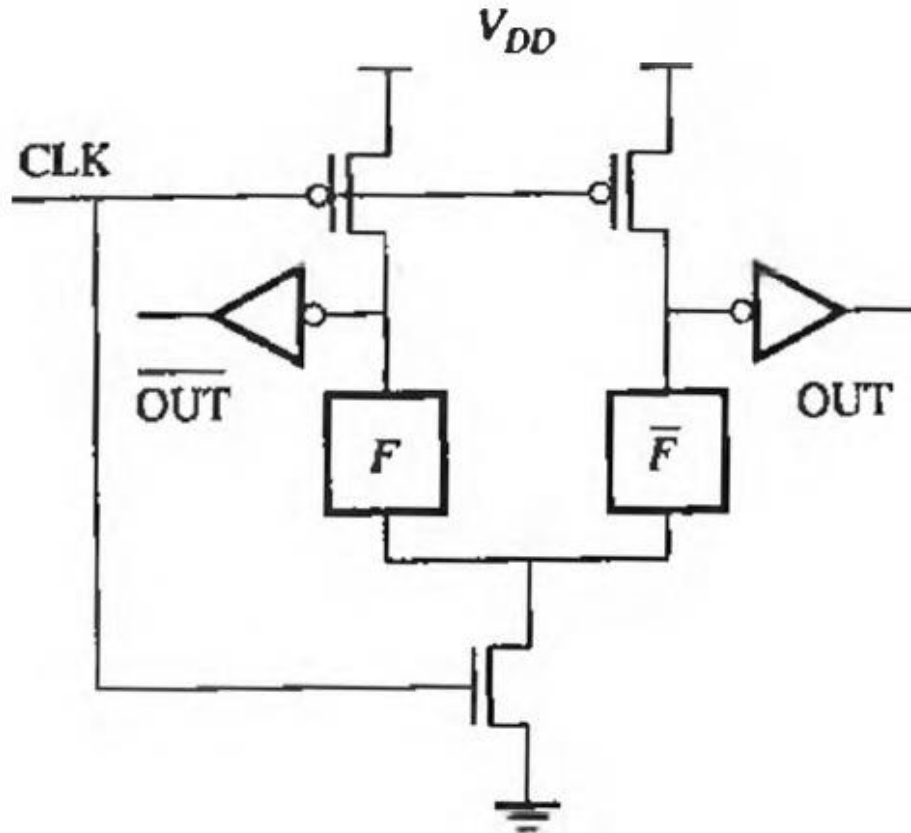
$$C_Y = (1 \text{ fF}/\mu\text{m})(16\lambda) = 16(0.05) = 0.8 \text{ fF}$$

Therefore, the worst-case charge sharing is given by

$$V^* = \frac{C_X(1.2 \text{ V})}{C_X + C_Y} = \frac{5(1.2 \text{ V})}{0.8 + 5} \approx 1.0 \text{ V}$$

This level is not possible at node Y. Set $V_Y = V_{DD} - V_{TN} = 0.73 \text{ V}$. The rest of the charge remains at node X. The remaining charge is $(5 \text{ fF})(1.2 \text{ V}) - 0.8 \text{ fF}(0.73 \text{ V}) = 5.42 \text{ fC}$. This translates to a voltage of 1.08 V at node X.

Dual-Rail (Differential) Domino Logic



Dual-Rail (Differential) Domino Logic

